

EDUCATION

Manipal Institute of Technology <i>Bachelor of Technology in Information Technology</i> ◦ Relevant Coursework: Operating Systems, Data Structures, Analysis of Algorithms, Computer Architecture, Machine Learning, Networking, and Databases	Manipal, India <i>August 2024 – June 2028</i>
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SKILLS SUMMARY

Languages: Go, Python, C/C++, JavaScript, TypeScript, HTML/CSS, SQL, Bash, Java
Frameworks: Node.js, Express.js, React.js, Next.js, Playwright, Puppeteer
Cybersecurity: Binary and Kernel Security, Reverse Engineering, IDA, Ghidra, GDB, x64dbg, Burp Suite
Tools: Docker, Git, PostgreSQL, MySQL, SQLite, MongoDB
Platforms: Linux, Web, Windows, Azure

EXPERIENCE

NielsenIQ <i>AntiBot Reverse Intern</i>	Pune, India <i>October 2026 – Upcoming</i>
Guestasy <i>SDE Intern</i> ◦ Google Reviews Scraper: Built a Playwright-based Google Maps scraper with stealth browser contexts, CAPTCHA detection, proxy retries, and incremental review collection. Technologies: Python, Playwright, JavaScript ◦ Booking.com Reviews Scraper: Developed a GraphQL-first review scraper with metadata resolution, proxy handling, and a Playwright-based HTML fallback. Technologies: Python, aiohttp, Playwright, GraphQL ◦ MakeMyTrip Reviews Scraper: Implemented an API-based scraper using Chrome TLS impersonation, geo-aware request headers, and proxy rotation for blocked requests. Technologies: Python, curl-cffi, aiohttp	Remote, Malaysia <i>June 2026 – August 2026</i>

Gephyr Private Limited [Certificate] <i>SDE Intern</i> ◦ Automated Boarding Pass Check-In System: Engineered an automated boarding pass check-in system using Puppeteer to authenticate users and complete check-ins, upload generated PDFs to Azure Blob Storage, and deliver them through a React-based frontend dashboard. Technologies: Puppeteer, Azure, React ◦ Search Optimizations: Reduced Hotel Search API response time from approximately 20 seconds to under 5 seconds by implementing database indexing, query optimization, result pagination, frequent-lookup caching, and selective field projection. Technologies: MongoDB, Node.js, React.js ◦ Automated Cancellation and Refund Module: Built a cancellation and refund module integrating Razorpay’s API, handling secure refund execution and atomically updating financial ledger records for hotel and bus microservices. Technologies: MongoDB, Node.js, React.js ◦ Multi-Provider Hotel API Integration: Integrated the Benzy-Akbar Hotel API with an existing provider, streamlined internal business logic, and improved combined search throughput by consolidating provider responses. Technologies: Node.js, MongoDB	Remote <i>June 2025 – June 2026</i>
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Cryptonite – Major Cybersecurity Student Project - 2nd Place Indian CTF team <i>Binary and Kernel Exploitation Head</i> ◦ Binary Exploitation: Conducted vulnerability research involving stack exploitation, heap exploitation, File Stream Oriented Programming, and memory corruption in the V8 engine ◦ Kernel Exploitation: Worked on virtual-machine exploitation and developed proof-of-concept exploits for privilege-escalation CVEs. Led the kernel and binary exploitation teams. ◦ Web Exploitation: Worked with Burp Suite and first-principles techniques involving XSS, SQL injection, XS-Leaks, race-condition vulnerabilities, SSTI, and related web exploitation methods.	Manipal, India
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PROJECTS

- **ropfuscator – ROP-Based Binary Obfuscator:** Implemented a technique to obfuscate ELF binaries using Return-Oriented Programming (ROP), inspired by a research paper. Added an option to parameterize the extent of obfuscation.
Tech: C/C++
- **ROP++ – ROP Gadget State-Space Solver:** Built a C++ state-machine-based ROP gadget solver that models register transformations as state transitions and searches for gadget chains to transform a target register from an initial value/state to a desired one. Implemented DFS and BFS search strategies for exploring candidate chains.
Tech: C++
- **Envv – Version Control for .env Files:** Designed an open-source, self-hostable CLI tool in Go that enables teams to manage .env files using Git-like commands such as `init`, `update`, and `import`.
Tech: Go, MongoDB

HONORS AND AWARDS

- **inCTF 2026** – Secured 1st place at inCTF, hosted by team bi0s, winning INR 1,00,000.
- **TexSAW 2026** – Secured 1st place in an accredited international CTF conducted by UT Dallas; contributed to clearing the binary exploitation category.
- **KaalChakra 2026** – Secured 2nd place in a national-level CTF conducted by NFSU Goa, winning INR 60,000.
- **CitadelCTF 2025** – Secured 1st place among more than 200 teams, winning INR 20,000.
- **R3ctf Writeup** – Won 300 dollars for a writeup on Escaping CET protection in CPUs
- **NitK ctf** – 2nd place, won a raspberry PI